

LiFePO4 Battery Specification

Model::TB4825F-T110D-3201-ESS



SHENZHEN TOPBAND BATTERY.,LTD

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V00	2021.03.16	First release	Ziqi Zhong

1. General Information

This specification defines the performance of rechargeable LiFePO₄ battery pack 48V25Ah manufactured by SHENZHEN TOPBAND BATTERY CO.,LTD describes the type, performance, technical characteristics, warning and caution of the battery pack.

2. Specification(@Battery initial Temp25±5℃)

NO.	Items	Criteria	Remarks
2.1	Rated Capacity	25Ah	@0.33C
2.2	Energy	1.2kWh	
2.3	Nominal Voltage	48V	
2.4	Outgoing Voltage	≥40.5V	
2.5	Internal resistance	≤40mΩ	
2.6	Communication	Can/RS485	
2.7	Limited charge voltage	52.5V	
2.8	Floating charge voltage	51V	
2.9	Maximum charge current	25A	
2.10	Maximum discharge current	25A	
2.11	Discharge cut-off voltage	40.5V	
2.12	IP rating	IP20	
2.13	Dimension	Width: 442±2mm	Install in 19" cabinet.
		Height: 88±2mm	
		Depth: 243±2mm	
2.14	Net weight	Approx:14.8 kg	
2.15	Battery case	white	
2.16	Operating Temperature	Charging: 0~50℃	

		Discharging: -20~60℃	
		Recommended operating temperature: 15℃~35℃	
2.17	Self-discharge rate	Residual capacity: ≤2%/month;≤12%/years	
2.18	Storage Temperature & Humidity Range	Less than 1 month: -20℃~45℃ , 45%RH~80%RH	
		Recommended storage environment: 15℃~35℃ , 45%RH~80%RH	
Long time storage: When the battery needs to be stored for a long time, it should be charged to a state of nearly 50% SOC with a voltage of about 48v and placed under the recommended storage conditions. Do at least one full charge and discharge cycle every 6 months (charge first, discharge and charge 50%).			

3. Product Performance

3.1 Standard Test Conditions

3.1.1 Unless otherwise specified, all performance tests are required conducted at temperature $25^{\circ}\text{C}\pm 2^{\circ}\text{C}$, Humidity less than 45%~75%RH.

3.1.2 Unless otherwise specified, the tested product is required unused within one month after outgoing.

NO.	Items	Criteria		Testing method
3.1	Rated Capacity	25Ah		@0.33C
3.2	Internal resistance	≤40mΩ		50% battery SOC state frequency of 1 KHZ ac resistance tester
3.3	Cycle life	≥4000cycle		@DOD80% @0.33C
3.4	Discharge Temperature Characteristics	-20℃	≥70%	At 25±5℃ discharge the battery with the current of 0.33C to the cut-off voltage and record charge capacity. Store the battery at various temperatures for 2h and discharge the battery with 0.33C to the cut-off voltage.
		-0℃	≥80%	
		25℃	≥100%	
		55℃	≥95%	
3.5	Charge Retention ability	Residual capacity≥90%		Charge the battery to full capacity and store it for 28days, and then discharge it with 0.33C to the cut-off voltage.
		Recovery capacity≥95%		

4. Protective circuit specification

The Battery Management System (BMS) can monitor and optimized each single prismatic cell during charge & discharge, to protect the battery pack overcharge, over discharge, short circuit. Overall, the BMS helps to ensure safe and accurate running.

No	Items	Description
Battery Alarm &Protection		
1	Over charge	Over-charge alarm for each cell
		3.65±0.03V
		Over-charge protection for each cell
		3.75±0.03V
		Over-charge release for each cell
		3.4±0.03V
		Over-charge alarm for total voltage
2	Over discharge	54.75V±0.5V
		Over-charge protection for total voltage
		56.25V±0.5V
		Over-charge release for total voltage
		54V±0.5V
		Over-charge release method
		Under the release voltage
3	Over current	Over-discharge alarm for each cell
		2.90±0.03V
		Over-discharge protection for each cell
		2.70±0.03V
		Over-discharge release for each cell
		3.00±0.03V
		Over-discharge alarm for total voltage
4	Over discharge	43.5V±0.5V
		Over-discharge protection for total voltage
		40.5V±0.5V
		Over-discharge release for total voltage
		45V±0.5V
		Over-discharge release method
		Charge to recovery
5	Over current	Charge over current alarm
		28±5A
		Charge over current protection
		30±5A
		Protection delay time
		5±1s
		Charge over current release method
6	Over current	Auto release after 1min;
		Discharge over current alarm
		25±5A
		Discharge over current protection
		30±10A
		Protection delay time
		5±1s

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		Over current release method	Auto release after 1min
4	Short circuit		280A, delay time 200~800us
5	Charge over temperature	Alarm @ $50\pm 3^{\circ}\text{C}$	
		Protect @ $55\pm 3^{\circ}\text{C}$	
		Release @ $45\pm 3^{\circ}\text{C}$	
6	Discharge over temperature	Alarm @ $60\pm 3^{\circ}\text{C}$	
		Protection @ $65\pm 3^{\circ}\text{C}$	
		Release @ $55\pm 3^{\circ}\text{C}$	
7	Charge low temperature protection	Alarm @ $3\pm 3^{\circ}\text{C}$	
		Protect @ $0\pm 3^{\circ}\text{C}$	
		Release @ $5\pm 3^{\circ}\text{C}$	
8	SOC	LOW SOC Alarm	5%
Monitor & Communication			
9	Communication	RS485	
10	Series &Parallel connection	No series connection; Support max. 14pcs in parallel.	

5.Equipment Interface Instruction

This section details the front panel of the interface functions

TB4825F product Front Interface



Items	Instruction
RUN	green LED flashing to show the battery running status.
ALM	Red LED flashing to show the battery has alarm, and lighting to show the battery is under protection.
SOC	4 green LEDs to show the battery' s current capacity.
RESET	Press for more than 3 seconds to activate or reset the battery pack
ADDR	4 ADD switches used by default,
RS485	Inverter communication interface or battery host computer software communication interface

ADD	Dial Switch				Instruction
/	#1	#2	#3	#4	
0	OFF	OFF	OFF	OFF	Pack0
1	ON	OFF	OFF	OFF	Pack1
2	OFF	ON	OFF	OFF	Pack2
3	ON	ON	OFF	OFF	Pack3
4	OFF	OFF	ON	OFF	Pack4
5	ON	OFF	ON	OFF	Pack5
6	OFF	ON	ON	OFF	Pack6
7	ON	ON	ON	OFF	Pack7
8	OFF	OFF	OFF	ON	Pack8
9	ON	OFF	OFF	ON	Pack9
10	OFF	ON	OFF	ON	Pack10

11	ON	ON	OFF	ON	Pack11
12	OFF	OFF	ON	ON	Pack12
13	ON	OFF	ON	ON	Pack13
14	OFF	ON	ON	ON	Pack14
15	ON	ON	ON	ON	Pack15

5.1 Pin definition

No.	RS485 Definition
1	B
2	A
3	GND
4	NC
5	NC
6	GND
7	A
8	B

6. Instructions

6.1 Transportation

- ✧ The battery need to be charged every 6 months if out of use
- ✧ No fall down, no pile up over 6 layers, and keep face up.

6.2 Install instructions

- ✧ Install battery & control module in standard 19" cabinet with 4 pcs M6 screws.

7. Warning & Tips

- ✧ Please read and follow the specification and caution remarks on battery surface before use the battery. Improper use may cause heat, fire, rupture, and damage or capacity deterioration of the battery. SHENZHEN TOPBAND BATTERY CO.,LTD. Describes is not responsible for any accidents caused by the usage without following our specification.
- ✧ The battery must be far away from heat source, high voltage, and avoid to be exposed in sunshine for long time.
- ✧ Do not put the battery in a charger or equipment with wrong terminals connected.
- ✧ Never connect the positive and negative of battery with metal.
- ✧ Avoid excessive physical shock or vibration. don't hit, fall, stamp on the battery
- ✧ Without the permission of the manufacturer and guidance, forbidden to remove or to assemble the battery
- ✧ Do not use the battery mixed with other different manufacturer, type, or model batteries.
- ✧ Keep the battery against high temperature. Otherwise it will cause battery heat, get into fire or lose some function and reduce the life.
- ✧ When battery run out of power, please charge your battery timely (≤ 15 day).
- ✧ Please use the matched or suggested charger for this battery.
- ✧ If battery emits peculiar smell, heating, distortion or appears any abnormity during working or storage, please stop using and take it out from device.
- ✧ If the battery leaks and get into the eyes or skin, do not wipe, instead, rinse it with clean water and see doctor immediately.
- ✧ Please far away from children or pets.
- ✧ Do not put disuse battery into a fire or water.

8. Battery operation instruction

8.1.1 Do not surpass the largest charging current that specification stipulated.

8.1.2 Do not surpass the highest limited voltage that specification stipulated.

8.1.3 Charging temperature: within temperature scope that specification stipulated。

8.1.4 Special note:

Short time doesn't affect the use of the battery overcharge too, but for a long period of time over discharge or over charge can affect the function of the battery failure, or the battery can't use permanent, appear serious safety hazards, and need long time floating please use the recommended floating model specification.

9. Other Chemical Reaction

Because batteries utilize a chemical reaction, battery performance will deteriorate over time even if stored for a long period of time without being used. In addition, if the various usage conditions such as charge, discharge, ambient temperature, etc. are not maintained within the specified ranges, the life expectancy of the battery may be shortened or the device in which the battery is used may be damaged by electrolyte leakage. If the discharge time is much shorter than the normal after full charged, even battery is charged correctly, and this may indicate it is time to change the battery.